WEEK 7 EXAM PRACTICE

Self-grading sheet, Question 1

Name _____ Date _____

Dantrolene blocks the ryanodine receptors in skeletal muscle. A motor neuron fires one action potential to a muscle fiber treated with dantrolene. Trace what happens from the neuromuscular junction to the crossbridges.

A. No end plate potential forms, because dantrolene blocks the nicotinic receptors, so the fiber never fires an action potential

B. The sarcoplasmic reticulum releases Ca2+ normally, but the Ca2+ cannot bind troponin, so no crossbridges form

C. Ca2+ flows in from the fluid outside the fiber through the DHP receptor, so the fiber contracts with normal force

D. The muscle action potential runs down the T tubule normally, but the sarcoplasmic reticulum releases little or no Ca2+, so tropomyosin stays over the binding sites and little or no force develops

The answer is D. Everything up to the T tubule is normal: acetylcholine opens the nicotinic receptors, the end plate potential reaches threshold, and the muscle action potential runs into the fiber. The DHP receptor senses it and tugs on the ryanodine receptor, but with the ryanodine receptor blocked, little or no Ca2+ leaves the sarcoplasmic reticulum. Troponin stays empty, tropomyosin stays over the myosin binding sites, and crossbridges cannot form. This is why dantrolene is the treatment for malignant hyperthermia, where the ryanodine receptor releases too much Ca2+.

Be honest with yourself. A point counts only if you clearly drew it or said it. If you are not sure you did, it does not count.

Your model: check each point you clearly drew or labeled (6 points)

- [ ] The junction in order: action potential at the axon terminal, Ca2+ in, acetylcholine released, nicotinic receptors opened, end plate potential, muscle action potential.
- [ ] The T tubule carries the action potential deep into the fiber, and the DHP receptor in its membrane changes shape.
- [ ] In skeletal muscle the DHP receptor is linked to the ryanodine receptor in the sarcoplasmic reticulum, so it pulls the channel open; no Ca2+ from outside the cell is needed.
- [ ] Normally Ca2+ leaves the sarcoplasmic reticulum, binds troponin, and tropomyosin rolls off the myosin binding sites on actin.
- [ ] The block is placed at the ryanodine receptor: little or no Ca2+ is released, so troponin stays empty and tropomyosin stays in place.
- [ ] Crossbridges cannot form, so little or no force develops, even though every electrical signal was normal.

Your teaching: check each point you clearly said (9 points)

- [ ] Read the question and said what it asks.
- [ ] Walked the neuromuscular junction from the motor neuron to the muscle action potential.
- [ ] Explained what the T tubule and the DHP receptor do.
- [ ] Explained how the DHP receptor opens the ryanodine receptor, and where the Ca2+ comes from.
- [ ] Traced the block from the ryanodine receptor to troponin, tropomyosin and the crossbridges.
- [ ] A is wrong: dantrolene acts inside the fiber, at the sarcoplasmic reticulum. The junction and the muscle action potential are normal.
- [ ] B is wrong: troponin can still bind Ca2+. The problem is that the Ca2+ is never released.
- [ ] C is wrong: skeletal muscle does not rely on Ca2+ entering from outside to start contraction. The DHP receptor works as a voltage sensor that pulls the ryanodine receptor open. Relying on Ca2+ entry is how heart muscle works.
- [ ] Gave the answer, D, and stated it again at the end.

Model points ___ of 6	Teaching points ___ of 9	Answer right? Yes / No	Your percent ___

Your score: add your model and teaching points, divide by 15, multiply by 75, and add 25 if your answer was right. Example: 5 model points and 7 teaching points is 12 of 15, which is 60; with the right answer, 85 percent.

No video? Grade your model from the photo of your whiteboard and leave the teaching points blank. Your score is then your model points divided by 6, times 75, plus 25 if your answer was right.

WEEK 7 EXAM PRACTICE

Self-grading sheet, Question 2

Name _____ Date _____

A patient's right leg is weak. When you tap below the right knee, the leg barely moves. The leg feels floppy when you move it, the thigh muscles have visibly shrunk, and small twitches ripple under the skin at rest. Stroking the sole does not make the big toe go up. Where is the problem most likely to be?

A. The lower motor neurons to the right thigh: their cell bodies in the ventral horn of the lumbar spinal cord, or their axons in the nerve
B. The left primary motor cortex
C. The right half of the spinal cord in the middle of the back, above the lumbar segments
D. The right cerebellum

The answer is A. Every finding belongs to lower motor neuron damage: weakness with low tone, a weak or absent reflex, wasting, twitches at rest (fasciculations) and no Babinski sign. The lower motor neuron is the last link to the muscle and part of the reflex arc, so when it is lost the reflex goes with it and the muscle loses its nerve supply. Lower motor neurons serve the muscles on their own side, so a right leg problem points to the right ventral horn or the right nerve.

Be honest with yourself. A point counts only if you clearly drew it or said it. If you are not sure you did, it does not count.

Your model: check each point you clearly drew or labeled (6 points)

- [ ] The upper motor neuron runs from the primary motor cortex, crosses at the bottom of the medulla, and descends in the corticospinal tract to the ventral horn.
- [ ] The lower motor neuron has its cell body in the ventral horn and runs out through the ventral root and the nerve to the muscle. It is the final common path and part of the reflex arc.
- [ ] Upper motor neuron damage: weakness, raised tone, brisk reflexes, a Babinski sign, and little wasting early on.
- [ ] Lower motor neuron damage: weakness, low tone, weak or absent reflexes, marked wasting, and fasciculations.
- [ ] Each of this patient's findings is placed in the lower motor neuron column.
- [ ] The side is placed: lower motor neurons serve muscles on their own side, so the problem is in the right ventral horn or the right nerve.

Your teaching: check each point you clearly said (9 points)

- [ ] Read the question and said what it asks.
- [ ] Walked the corticospinal pathway from the cortex to the ventral horn.
- [ ] Explained what the lower motor neuron is and why the knee reflex runs through it.
- [ ] Matched each finding to upper or lower motor neuron damage, with a reason.
- [ ] Explained why the muscle wastes and twitches when it loses its nerve supply.
- [ ] B is wrong: damage to the left motor cortex is upper motor neuron damage, which gives brisk reflexes, raised tone, a Babinski sign and little early wasting.
- [ ] C is wrong: a cut through the cord above the lumbar segments interrupts the upper motor neurons to the leg, so it gives upper motor neuron signs below the cut.
- [ ] D is wrong: cerebellar damage makes movement clumsy and poorly aimed on the same side, but it does not waste the muscle or cause fasciculations.
- [ ] Gave the answer, A, and stated it again at the end.

Model points ___ of 6	Teaching points ___ of 9	Answer right? Yes / No	Your percent ___

Your score: add your model and teaching points, divide by 15, multiply by 75, and add 25 if your answer was right. Example: 5 model points and 7 teaching points is 12 of 15, which is 60; with the right answer, 85 percent.

No video? Grade your model from the photo of your whiteboard and leave the teaching points blank. Your score is then your model points divided by 6, times 75, plus 25 if your answer was right.